

Initial ODMA-aware decoding

The structured signal model, BP/EP derivation, and V1-V4 development

Stage	1 of 5
Updated	5 September 2026
Status	Historical stage reconstructed from the original derivations and retained implementation

This report is a chronological research record. Measured results, interpretations, and open hypotheses are labelled separately. Detailed numerical ledgers remain in **results/**.

1 Position in the research story

The project began with a receiver-side question: if every message waveform has explicit on-off division multiple access (ODMA) structure, can a decoder exploit the resulting sparse factor graph more effectively than a generic global sparse-recovery method? This report retains the algebra used to answer that question. It is intentionally more detailed than a present-day summary because these equations explain how the early algorithms were constructed and why their failure modes led to the later codebook-design programme.

The central receiver was a two-stage hybrid:

1. a linear-Gaussian resource stage, which compresses interfering contributions to means and covariances;
2. a discrete block stage, which enumerates valid local count vectors and uses the exact local codebook.

The stages exchange extrinsic Gaussian information. Successive V1–V4 variants changed the observation model, covariance approximation, schedule, and fading treatment; they did not change the underlying unsourced target.

Evidence status: Historical reconstruction. The factorisation and current Graph-BP implementation are directly verifiable; some superseded V1–V4 variants survive only as research records and should not be treated as current baselines.

2 ODMA-URA signal model

2.1 Patterns, local codebooks, and global messages

Let n be the number of channel uses and let Q be the number of ODMA placement patterns. Pattern q occupies d_q resources $S_q \subseteq \{1, \dots, n\}$ and is represented by

$$P_q \in \{0, 1\}^{n \times d_q}, \quad P_q^H P_q = I_{d_q}.$$

If $x_q \in \mathbb{C}^{d_q}$ is a local signal, $P_q x_q$ embeds it into the full resource grid. Pattern q owns V_q local codewords, stored as rows of

$$C_q \in \mathbb{C}^{V_q \times d_q}, \quad c_{q,v} = C_q[v, :]^T.$$

The global message (q, v) transmits

$$\phi_{q,v} = P_q c_{q,v} \in \mathbb{C}^n.$$

When every pair is legal, the total message alphabet is $M = \sum_q V_q$ and the explicit global codebook is

$$\Phi = [P_1 C_1^T \mid P_2 C_2^T \mid \dots \mid P_Q C_Q^T] \in \mathbb{C}^{n \times M}.$$

Column normalisation $\|\phi_m\|_2^2 = 1$ fixes the transmitted energy of every message.

The B payload bits select one of $M = 2^B$ columns. They are labels, not binary channel symbols: the entries of ϕ_m may be real or complex amplitudes, so each physical channel use may carry magnitude and phase.

2.2 Unsourced count target

For K_a active devices, define

$$a_q \in \mathbb{Z}_+^{V_q}, \quad a_{q,v} = \text{number of devices that chose } (q, v).$$

The local superposition in block q and the global resource signal are

$$x_q = C_q^T a_q \in \mathbb{C}^{d_q}, \quad s = \sum_{q=1}^Q P_q x_q = \Phi a, \quad \mathbf{1}^T a = K_a.$$

The decoder seeks the multiset a , not device identities. If two devices send the same message then $a_m = 2$; this is a valid message collision, not two separately identifiable users.

2.3 Observation models used during V1–V4

The original sequence can be written as increasingly difficult versions of the same resource signal.

V1: scalar, known unit channel.

$$y = s + z = \sum_q P_q C_q^T a_q + z, \quad z \sim \mathcal{CN}(0, \sigma^2 I_n).$$

V2: multiple antennas with a known common signature. For M_{ant} receive antennas and known $h \in \mathbb{C}^{M_{\text{ant}}}$,

$$Y = s h^T + Z, \quad Z_{i,:} \sim \mathcal{CN}(0, \sigma^2 I).$$

Let $\gamma = \|h\|_2^2$. Matched combining gives the sufficient scalar observation

$$\bar{y} = \frac{Y h^*}{\gamma} = s + \bar{z}, \quad \bar{z} \sim \mathcal{CN}(0, \sigma^2 \gamma^{-1} I_n).$$

In the implementation $h = \mathbf{1}$, so $\gamma = M_{\text{ant}}$. The orthogonal antenna projector

$$Q_{\perp} = I - \frac{h h^H}{\gamma}$$

contains no signal under this model and can estimate noise because $Q_{\perp} Y_i = Q_{\perp} Z_i$.

V3: message-dependent unknown fading. With channel vector h_u for device u and selected message m_u ,

$$Y = \sum_{u=1}^{K_a} \phi_{m_u} h_u^T + Z.$$

Now activity and channel are bilinearly coupled. A count a_m alone is no longer sufficient unless the fading coefficients are known, integrated out, or replaced by an energy/support statistic.

V4: exact small-state reference. At tiny dimensions, the intended posterior over discrete counts and any retained channel state can be enumerated or integrated exactly. This is a correctness reference, not a scalable receiver.

3 Factor graph in block variables

The early decoder used one variable node per ODMA block signal x_q and one factor per resource. If coordinate j of block q is placed at resource i , then $P_q[i, j] = 1$. Define the neighbours

$$\mathcal{N}(i) = \{(q, j) : P_q[i, j] = 1\}, \quad \mathcal{N}(q) = S_q.$$

Under V2, resource i obeys

$$Y_i = h \sum_{(q,j) \in \mathcal{N}(i)} x_{q,j} + Z_i.$$

The exact posterior factors into resource likelihoods and discrete block priors, but shared resources create loops. The algorithm therefore approximates every incoming block-coordinate belief by a Gaussian.

3.1 Stage 1: linear-Gaussian resource update

Suppose edge $(q, j) \rightarrow i$ carries

$$x_{q,j} \sim \mathcal{CN}(m_{qj \rightarrow i}, v_{qj \rightarrow i}).$$

Excluding the candidate edge, the other contributions are approximated by

$$\mu_{i \setminus qj} = \sum_{(b,k) \in \mathcal{N}(i) \setminus (q,j)} m_{bk \rightarrow i}, \quad \nu_{i \setminus qj} = \sum_{(b,k) \in \mathcal{N}(i) \setminus (q,j)} v_{bk \rightarrow i}.$$

After matched combining, the extrinsic scalar observation is

$$r_{i \rightarrow qj} = \bar{y}_i - \mu_{i \setminus qj} = x_{q,j} + \epsilon_{i \rightarrow qj}, \quad \text{Var}(\epsilon_{i \rightarrow qj}) = \nu_{i \setminus qj} + \sigma^2/\gamma.$$

For vector block updates, stacking the relevant coordinates gives a Gaussian cavity

$$q^{\setminus q}(x_q) = \mathcal{CN}(x_q; r_q^{\text{cav}}, V_q^{\text{cav}}).$$

3.2 Stage 2: exact discrete block update

The structural constraint is

$$x_q = C_q^T a_q, \text{quad} a_q \in \mathcal{A}_q \subset \mathbb{Z}_+^{V_q}.$$

For a Poisson approximation with rate λ per message,

$$p(a_q) = \prod_{v=1}^{V_q} e^{-\lambda} \frac{\lambda^{a_{q,v}}}{a_{q,v}!}.$$

Given the Gaussian cavity, the posterior weight of state $a \in \mathcal{A}_q$ is

$$\omega_q(a) \propto p(a) \exp \left[-(C_q^T a - r_q^{\text{cav}})^H (V_q^{\text{cav}})^{-1} (C_q^T a - r_q^{\text{cav}}) \right].$$

Normalising these weights yields the exact moments inside the enumerated local state set:

$$\begin{aligned} \hat{a}_q &= \sum_{a \in \mathcal{A}_q} \omega_q(a) a, \\ \hat{x}_q &= \sum_{a \in \mathcal{A}_q} \omega_q(a) C_q^T a, \\ \hat{\Sigma}_q &= \sum_{a \in \mathcal{A}_q} \omega_q(a) (C_q^T a - \hat{x}_q)(C_q^T a - \hat{x}_q)^H. \end{aligned}$$

The block MAP state is $a_q^{\text{MAP}} = \arg \max_{a \in \mathcal{A}_q} \omega_q(a)$.

The state count explains the cost. If at most $k_{\max}$ local messages are nonzero and each count is at most $c_{\max}$, then

$$|\mathcal{A}_q| = \sum_{k=0}^{k_{\max}} \binom{V_q}{k} c_{\max}^k.$$

Truncation makes the decoder executable, but it also changes the inference domain.

3.3 Extrinsic EP site update

Write a Gaussian in natural parameters $(\Lambda, \eta) = (\Sigma^{-1}, \Sigma^{-1}\mu)$. EP divides the moment-matched posterior by the cavity:

$$\Lambda_q^{\text{site}} = \hat{\Sigma}_q^{-1} - (V_q^{\text{cav}})^{-1}, \quad \eta_q^{\text{site}} = \hat{\Sigma}_q^{-1} \hat{x}_q - (V_q^{\text{cav}})^{-1} r_q^{\text{cav}}.$$

These site parameters are damped and returned to the linear system. The current implementation forms the cavity by omitting the old block site from the global precision system rather than subtracting nearly equal posterior and site parameters after solving.

3.4 EM updates used by the approximation

If $M = \sum_q V_q$, the Poisson-rate update is

$$\lambda^+ = \frac{1}{M} \sum_q \mathbf{1}^T \hat{a}_q.$$

Let $\hat{y} = \sum_q P_q \hat{x}_q$. A moment-corrected noise update has the form

$$(\sigma^2)^+ = \frac{\|Y - \hat{y}h^T\|_F^2 + \text{gamma} \sum_q \text{tr}(\hat{\Sigma}_q)}{nM_{\text{ant}}}.$$

These are empirical-Bayes updates inside an approximate posterior, not ground-truth parameter estimates.

4 Implementation sequence and what each change meant

Variant	Structural change	Intended diagnostic
V1	Scalar, no-fading resource messages	Establish the two-stage algebra in the simplest channel
V2	Common known multi-antenna signature	Test antenna combining without bilinear channel uncertainty
V2b	Block/full covariance rather than diagonal moments	Measure the value of within-block correlations
V2c	Onsager-like self-response correction	Reduce feedback through loopy shared resources
V2d	Serial/turbo block schedule	Use freshly updated cavities rather than stale parallel messages
V2e	Global EP with cavity-by-omission	Avoid unstable subtraction when forming cavities
V2f	Fractional EP site power	Limit information injection when sites become overconfident
V3	Explicit latent fading variables	Represent the physical bilinear channel rather than pre-equalise it
V3a	Energy/support scoring inspired by noncoherent aggregation	Retain multi-antenna support evidence at lower cost
V4	Exact small-state likelihood/reference	Separate approximation failure from model/code errors

Fractional EP replaces a full site update by a powered one. In natural coordinates a simple damped form is

$$\Lambda_q^+ = (1 - \alpha)\Lambda_q^{\text{old}} + \alpha\beta\Lambda_q^{\text{site}}, \quad \eta_q^+ = (1 - \alpha)\eta_q^{\text{old}} + \alpha\beta\eta_q^{\text{site}}, \quad 0 < \beta \leq 1.$$

It can stabilise an iteration by shrinking the update, but does not correct an inaccurate local likelihood or remove short loops. Similarly, a serial schedule changes dynamics, not the fixed-point objective.

5 Why the graph decoder became unstable

The observed high-load failure has a concrete EP mechanism. When the discrete posterior places nearly all mass on one state, $\hat{\Sigma}_q$ becomes very small. Then $\hat{\Sigma}_q^{-1}$ and hence the site precision become large. That strong precision tightens the cavities of every overlapping block. With many short loops, the next blocks become overconfident in turn, and the feedback gain can exceed one. Damping and fractional powers slow this amplification; they do not guarantee that the Gaussian projection is accurate.

Four additional limitations remained:

1. the independent Poisson prior does not impose the exact multinomial constraint $\mathbf{1}^T a = K_a$;
2. local state caps may exclude the true high-occupancy block state;
3. ODMA reuses a small number of support subspaces, creating correlated columns and local ambiguities;
4. under unknown fading, activity and channel scale/phase are not separately identified without extra structure.

6 Baseline comparison and historical conclusion

The important baselines were global OMP/NNOMP, linear MMSE, SIC, AMP-like iterations, local BlockMAP, and exact tiny references. Their assumptions differ. In particular, NNOMP was usually supplied the true K_a and therefore received oracle knowledge of how much total mass to select. Exact BlockMAP ignored or only approximated cross-block interference. V4 was too small to be a practical competitor.

Qualitatively, the structured graph decoder could work in easy regimes but did not show a stable advantage as load, overlap, or fading uncertainty increased. Global NNOMP was simpler and usually stronger in the comparisons that drove the project. This did not establish that ODMA was information-theoretically inferior. It established that adding more local message-passing machinery was not isolating the dominant loss.

The next question was therefore causal: once the true active columns are supplied, are ODMA columns intrinsically hard to fit, or is the main loss the search for those columns? Report 2 places every construction in one global count model, catalogues the decoder objectives tried, and answers that question with an oracle-support control.